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Intensive Versus Conventional Insulin Therapy in Type 2 Diabetes Patients Undergoing D2 Gastrectomy for Gastric Cancer: A Randomized Controlled Trial

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Abstract

Background This study was to compare the effect of intensive insulin therapy (IIT) to conventional insulin therapy (CIT) on postoperative outcomes among type 2 diabetes mellitus (DM) patients who underwent D2 gastrectomy for gastric cancer.

Methods We randomly assigned gastric cancer patients with type 2 DM who underwent radical gastrectomy to receive IIT (maintenance of blood glucose at a level between 4.4 and 6.1 mmol/l) with insulin infusion or CIT (maintenance of blood glucose at a level between 10 and 11.1 mmol/l) during the postoperative period.

Results Of the 179 eligible patients, 92 patients were assigned to receive IIT and 87 patients to receive CIT. Mean blood glucose concentrations were lower in the intensive group (IG) than in the conventional group (CG) (5.5 ± 0.8 vs. 9.9 ± 1.0 mmol/l, $P < 0.001$). Hypoglycemia occurred in 6 patients (6.5%) in the IG ($P = 0.029$) versus in 1 patient (1.1%) in the CG. Hospital mortality did not differ significantly between two groups (4.3% vs. 5.7%, $P = 0.742$). However, IIT significantly reduced morbidity (from 18.4 to 7.6%, $P = 0.031$). Also, IIT shortened the days to suture removal, postoperative hospital stay, and postoperative duration of antibiotic use. The HOMA-IR score was lower at all time points in IG. Moreover, IIT increased the postoperative HLA-DR expression on monocytes on postoperative days 3 and 5.

Conclusions IIT significantly reduced short-term morbidity but not mortality among type 2 DM patients who underwent D2 gastrectomy for gastric cancer. Furthermore, a possible mechanism of suppression of the insulin resistance and improvement of HLA-DR expression may partially explain the benefits of IIT.

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Introduction

Increasing evidence shows that diabetes mellitus (DM) has many deleterious effects on patients after cardiac surgery, including a higher incidence of infection, ischemic events, neurologic and renal complications, and mortality [1–3]. Hyperglycemia has been identified as a risk factor and predictor for postoperative morbidity and mortality [4, 5]. Van der Berghe et al. [4] reported the use of exogenous insulin to maintain a blood glucose level no higher than 6.1 mmol/l reduced mortality and morbidity among critically ill patients in the surgical intensive care unit, regardless of whether they had a history of diabetes. In addition, in 2006, they found that strict normalization of blood glucose level (4.4–6.1 mmol/l), compared to conventional therapy (maintenance of blood glucose at 11.1 mmol/l or less), significantly reduced morbidity but not mortality among all the patients in the medical intensive care unit [6]. Moreover, several studies have demonstrated that intensive glucose control during cardiac surgery is associated with a decreased infection rate and improved survival [2, 3].

However, most recently, there has been controversy over the safety, increased workload, and benefits of IIT. A large, international, randomized trial showed that a blood

glucose target of less than 10 mmol/l resulted in lower mortality than a target of 4.5–6.0 mmol/l. The lower glucose target was not recommended in critically ill adults [7]. Nevertheless, the study cannot exclude the possibility that intensive glucose control may benefit some patients. Therefore, the appropriate glucose target and specific subpopulations that might benefit from IIT have yet to be identified [8]. Of note, to date, existing studies on perioperative glucose control focus mainly on diabetic or nondiabetic patients undergoing cardiac surgery. However, well-designed randomized studies on surgical homogeneous populations outside of cardiac surgery were lacking. Our previous study indicated that intensive insulin therapy improved the short-term morbidity in nondiabetic patients who undergo elective gastrectomy [9].

The impact of intensive glucose control on short-term mortality and morbidity in type 2 DM patients undergoing D2 gastrectomy for gastric cancer is unclear. Thus, we performed a prospective, randomized, controlled study to investigate that whether the type 2 DM patients who underwent radical gastrectomy for gastric cancer will benefit from IIT. Meanwhile, insulin sensitivity indicated as HOMA-IR score and HLA-DR expression on monocytes were examined to explore the potential mechanism of IIT. The primary outcome was postoperative short-term complication rate. The secondary outcomes included postoperative 28-day mortality rate, HOMA-IR score, and HLA-DR.

Patients and methods

Patients and design

The study was approved by the Ethics Committee of the hospital, and informed written consent was obtained from each patient. From January 2005 to December 2009, we studied 220 consecutive adult patients with type 2 DM who were to undergo open elective gastrectomy for gastric cancer. In all cases, diagnosis of type 2 diabetes and gastric cancer had been confirmed before surgery. Type 2 DM was defined according to 1999 World Health Organization criteria [10]. Exclusion criteria included the following: (1) age was less than 16 years; (2) severe obesity (body mass index [BMI] > 30 kg/m^2) or severe malnutrition (BMI < 15 kg/m^2); (3) expected SICU stay after surgery was less than 24 h; (4) tumor was unresectable or the patient with late-stage cancer underwent palliative surgery; (5) pregnancy; (6) took corticosteroids, steroids, growth hormone, or immunosuppressive drugs within 2 weeks prior to the study; (7) patient received neoadjuvant chemotherapy; (8) patient was diagnosed with gastric stump cancer or recurrent gastric cancer.

To avoid the serious harm of severe hypoglycemia, we decided to use a single-center, unblinded design. The continuous insulin infusion protocol used in this study was the Portland protocol [11]. The protocol was implemented by a multidisciplinary team that included well-trained surgeons, diabetologists, and SICU nurses who had extensive experience in blood glucose control and could promptly identify early hypoglycemia and give appropriate intervention. After complete preoperative assessment and preparation, the patients underwent the surgical procedure under general anesthesia. All the patients were admitted into SICU after surgery. Upon SICU admission, the eligible patients for this study were randomly divided into an intensive group (IG) or a conventional group (CG). Fifty IU of regular insulin in 50 ml of normal saline was administered using an infusion pump (Smiths Medical Instrument Co., Zhejiang, China). In IG treatment, the insulin infusion was started if the blood glucose levels exceeded 6.1 mmol/l and was adjusted to maintain the blood glucose target between 4.4 and 6.1 mmol/l. In CG treatment, the insulin infusion was started if the blood glucose level exceeded 12.0 mmol/l and was adjusted to maintain the blood glucose target between 10.0 and 11.0 mmol/l. A blood glucose level of 2.2 mmol/l or less was considered severe hypoglycemia.

Blood glucose levels were measured and monitored using a bedside glucometer (OneTouch Ultra 2, LifeScan) or laboratory analyzers in routine use. The glucometers were calibrated regularly by the manufacturer to ensure the accuracy and reliability. The infusions were continued postoperatively until oral intake or enteral nutrition was established, after which the usual treatment for DM was resumed. The postoperative mean glucose concentration was calculated as the composite average of all the glucose levels from the period immediately after surgery to the end of protocol.

The guidelines of the Japanese Research Society for the Study of Gastric Cancer were used in this study for standard surgical treatment and pathological assessment [12]. All eligible patients underwent elective radical gastrectomy (subtotal or total) with D2 lymphadenectomy. All the operations were performed by qualified, experienced surgeons in our hospital, a high-volume gastric cancer center.

Insulin sensitivity

In the present study, Homeostasis Model Assessment (HOMA) was chosen to assess insulin resistance (IR) because HOMA-IR strongly correlates with the euglycemic clamp IR in type 2 DM patients [13, 14]. Although the latter is considered the golden standard method for assessing insulin sensitivity, it is time-consuming, laborious, and costly. HOMA-IR can be reliably used in large-scale or epidemiologic studies and has shown good reproducibility

and consistency [14–16]. The HOMA-IR score is calculated using the following mathematic formula: HOMA-IR = FI × FG/22.5, where FI is fasting plasma insulin (μU/ml), FG is fasting plasma glucose (mmol/l), and 22.5 is a constant. In view of the pulsatile type of insulin secretion, the FI value was the mean of three results at 5-min intervals (0-, 5-, and 10-min samples). Insulin was tested using insulin reaction kits (Elecsys Automatic Electrochemiluminescence Immunoassay Instrument, Roche, Germany). Blood glucose was detected using an automatic chemistry analyzer (Hitachi, Japan).

The blood samples for FI and FG analysis were drawn from the median cubital vein the morning of preoperative day 1 after overnight fasting (baseline) and on postoperative days 1, 3, and 5 when fluid infusions were discontinued at least 6 h. FI and FG analyses were performed immediately. These blood samples were also used to measure the HLA-DR expression on monocytes. After centrifugation, samples were kept frozen −80°C until assay.

Assay of HLA-DR expression on monocytes

Blood sample collection was described as above. The HLA-DR expression on monocytes was detected by flow cytometry (BD FACS Aria) using fluorescein isothiocyanate-labeled anti-14 and phycoerythrin-labeled anti-HLA-DR (Multisciences Biotech Co., Shanghai, China).

Outcome measurements and definitions

Patients were followed from hospital admission to 28 days after surgery. Demographic and clinical characteristics and intraoperative data of each patient are presented in Table 1. The primary outcome was the postoperative short-term complication rate. Postoperative complications were recorded as wound infection, intra-abdominal infection, sepsis, urinary tract infection, or pneumonia. The wound was evaluated daily and 15 and 28 days after surgery by the same surgeon who was blinded to the treatment assignments. When pus could be expressed from the incision or aspirated from a loculated mass within the wound, wound infection was suspected. In this case, pus was cultured and wound infection was proved by a positive culture. All the wound infections diagnosed within 28 days after surgery were included in the data analysis. Intra-abdominal infection was defined as a positive culture from drainage or puncture fluid with a need for systemic antibiotic treatment and consistent with the image and clinical findings. The definition of sepsis was given previously [17]. The criteria for urinary tract infection required positive culture of urine. It was defined as pneumonia when cultures from sputum, pleural fluid, or empyema fluid were positive, consistent with the diagnosis and clinical symptoms. A chest radiograph diagnostic of pulmonary filtrates was also considered pneumonia. In addition, days to suture removal, postoperative length of hospital stay, and duration of antibiotic use were also recorded. Hospital mortality was defined as death from any cause within 28 days of the operation or during the period of hospital stay.

In our study, perioperative antibiotic prophylaxis was executed as follows: cefamandole 1 g/dose was administered intravenously 30 min before the initial surgical incision and then 1 g/8 h for 2 days postoperatively. No other antibiotic was given thereafter unless an established infection could be verified or the patient's clinical course highly suggested development of septic complications. If postoperative infection developed, appropriate aerobic and anaerobic cultures were taken from all sites of infections. Furthermore, antibiotic sensitivity of aerobic pathogens was determined by sensitivity disc testing in the microbiology laboratory. Patients with defined infection received empirical antibiotic therapy before the positive culture results were available. Subsequently, appropriate antibiotic therapy was given based on bacterial culture and sensitivity testing results. The antibiotics used in this study included ceftriaxone, ciprofloxacin, piperacillin-tazobactam, and meropenen. The site of infection, causative agent, and local resistance patterns were taken into account when selecting the antibiotic to use.

Statistical analysis

The expected postoperative complication rate of the conventional group was 25%. The minimum required sample size was determined by using an appropriate formula that would provide 80% power to detect a 16% difference in the postoperative complication rate at a 0.05 level (two-sided test). Approximately 86 patients were required to be recruited in each group to fulfill the aim of the study. A simple randomization method (300 random numbers were generated through a random number table) with concealment was used for allocating the patients to different groups in this prospective study. We needed only 172 patients to achieve our target. However, the main drawback of simple randomization is the possibility of imbalanced group sizes in small randomly controlled trials (RCTs). Therefore, we continued to recruit patients until each group had at least 86 patients, which could explain the unequal numbers in the two groups.

The quantitative variables were presented as mean ± SD or median. The qualitative variables were summarized as frequency and percentage. The results were reported separately for each of two groups. Differences between the two groups were analyzed with the χ^2 test (or Fisher's exact test, where appropriate) for comparison of proportions,

Table 1 Patients' characteristics and surgical data

Variables	IG ($n = 92$)	CG ($n = 87$)	P
Gender (f/m)	64/28	57/30	0.563
Age (years)	58.2 ± 6.3	59.4 ± 7.3	0.248
BMI (kg/m^2)	21.1 ± 2.0	22.2 ± 2.6	0.002
Current smoking [*n* (%)]	32(34.8)	33(37.9)	0.662
DM duration (years), median	5.5	6.0	0.250
HbA1c (%)	7.5 ± 0.7	7.3 ± 0.6	0.081
Fasting C-peptide (pmol/l)	933 ± 196	948 ± 170	0.579
FBG (mmol/l)	6.8 ± 0.6	7.0 ± 0.7	0.116
Treatment for DM [*n* (%)]			0.592
Insulin	60 (65.2)	63 (72.4)	
Oral agents	25 (27.2)	19 (21.8)	
None	7 (7.6)	5 (5.7)	
ASA (1/2/3) [*n* (%)]	25/50/17 (27.2/54.3/18.5)	21/52/14 (24.1/59.8/16.1)	0.764
Preop comorbidity [*n* (%)]	35 (38.0)	31 (35.6)	0.738
Hypertension	17 (18.5)	15 (17.2)	
Coronary artery disease	2 (2.2)	1 (1.1)	
Cardiac insufficiency	7 (7.6)	5 (5.7)	
Pulmonary disease	4 (4.3)	5 (5.7)	
Renal insufficicency	1 (1.0)	2 (2.3)	
Liver insufficiency	4 (4.3)	3 (3.4)	
Type of operation [*n* (%)]			0.839
Distal gastrectomy	59 (64.1)	53 (60.9)	
Proximal gastrectomy	13 (14.1)	15 (17.2)	
Total gastrectomy	20 (21.7)	19 (21.8)	0.732
Combined resection [*n* (%)]			
Yes	11 (12.0)	9 (10.3)	
No	81 (88.0)	78 (89.7)	
Type of anastomosis [*n* (%)]			0.820
Billroth I	48 (52.2)	40 (46.0)	
Billroth II	11 (12.0)	13 (14.9)	
Roux-en-Y	20 (21.7)	19 (21.8)	
Billroth reverse	13 (14.1)	15 (17.2)	0.872
Location of tumor [*n* (%)]			
More than two-thirds of stomach	15 (16.3)	13 (14.9)	
Upper third	14 (15.2)	12 (13.8)	
Middle third	20 (21.7)	16 (18.4)	
Distal third	43 (46.7)	46 (52.9)	
Length of surgery (min)	145 ± 18	139 ± 16	0.021
Estimated blood loss (ml)	345 ± 65	360 ± 71	0.134
NNISS score (0/1/2) [*n* (%)]	16/55/21 (17.4/59.8/22.8)	14/50/23 (16.1/57.5/26.4)	0.851
SENIC score (1/2/3) [*n* (%)]	8/57/27 (8.7/62.0/29.3)	13/51/24 (14.8/58.0/27.3)	0.447

Data are given as mean ± SD, median, or number (%)

IG intensive group, *CG* conventional group, *BMI* body mass index, *DM* diabetes mellitus, *Hb1Ac* glycolsylated hemoglobin, *FBG* fasting blood glucose, *ASA* American Society of Anesthesiologists classification, *Preop* preoperative, *NNISS* National Nosocomial Infection Surveillance System, *SENIC* Center for Disease Control Study on the Efficacy of Nosocomial Infection Control

Student's *t* test for comparison of normally distributed data, and the Mann–Whitney *U* test for unnormally distributed data. The value $P < 0.05$ was considered statistically significant. The analyses were performed on an intent-to-treatment basis. Statistical analysis was performed using SPSS Advanced Statistical 13.0 software (SPSS Inc., Chicago, IL).

Results

Demographic and surgical data

Among 210 patients screened, 179 patients were eligible and formed the study population. Excluded from this study were patients undergoing palliative surgery ($n = 17$), patients

who accepted neoadjuvant chemotherapy ($n = 7$), patients who underwent D1 gastrectomy ($n = 5$), and patients diagnosed with gastric stump cancer or recurrent gastric cancer ($n = 2$). Finally, 92 patients were assigned into the intensive group (IG) and 87 patients into the conventional group (CG). The patients of the two groups underwent open elective radical subtotal or total gastrectomy with or without combined resection. The main patient characteristics and surgical data are reported in Table 1. Surgical procedures, type of anastomosis, duration of surgery, estimated blood loss, and SENIC and NNISS scores for the risk of infection were similar for the two groups. However, the BMI of the CG patients was greater than that of the IG patients (22.2 ± 2.6 vs. 21.1 ± 2.0, $P = 0.002$).

Outcome measurements

Postoperative mean blood glucose concentrations were lower in IG than in CG (5.5 ± 0.8 vs. 9.9 ± 1.0 mmol/l, $P < 0.001$). Severe hypoglycemia was recorded in 6 of the 92 patients (6.5%) in IG compared with 1 of 87 patients (1.1%) in CG ($P = 0.029$). Because of appropriate treatment, no sequelae of severe hypoglycemia were reported in the 7 patients.

Table 2 gives the postoperative short-term outcomes for the two groups. With respect to hospital mortality, 4 IG patients (4.3%) died versus 5 CG patients (5.7%). There was no significant difference in mortality rate ($P = 0.742$). The four deaths in the IG group were due to cerebrovascular accident ($n = 1$), acute massive pulmonary embolism ($n = 1$), excessive intra-abdominal bleeding ($n = 1$), and renal failure with severe sepsis ($n = 1$). The five deaths in the CG group were due to respiratory failure due to pulmonary infection ($n = 1$), acute cardiac failure ($n = 1$), acute massive pulmonary embolism ($n = 1$), acute renal failure due to severe intra-abdominal infection ($n = 1$), and uncontrolled gastrointestinal bleeding ($n = 1$). The results showed there was no close association between intensive insulin therapy and mortality rate.

Overall, intensive insulin therapy reduced the short-term morbidity rate from 18.4 to 7.6% ($P = 0.031$). When it comes to types of complications, the wound infection rate was 4.3% in IG vs. 13.8% in CG, intra-abdominal infection rate 2.2 vs. 10.3%, pneumonia rate 3.3 vs. 11.5%, urinary tract infection rate 4.3 vs. 3.4%, and sepsis rate 2.2 vs. 3.4%, respectively. Sutures were removed significantly later in the CG. Furthermore, the duration of postoperative stay and the length of postoperative antibiotic use were evaluated; both had significant difference between the two groups (Table 2).

Table 2 Outcomes after surgery in two groups

Variables	IG ($n = 92$)	CG ($n = 87$)	P
Postop hospital mortality [n (%)]	4 (4.3)	5 (5.7)	0.742
Postop complications (total) [n (%)]	7 (7.6)	16 (18.4)	0.031
Type of complications [n (%)]			
Wound infection	4 (4.3)	12 (13.8)	
Intra-abdominal infection	2 (2.2)	9 (10.3)	
Pneumonia	3 (3.3)	10 (11.5)	
Urinary tract infection	4 (4.3)	3 (3.4)	
Sepsis	2 (2.2)	3 (3.4)	
Days to suture removal (median)	8	9	<0.001
Postop hospital stay (days) (median)	8	10	0.002
Postop duration of antibiotic use (days) (median)	3	4	<0.001

Data are given as number (%) or median. Some patients had more than one complication

Postop postoperative, *SICU* surgical intensive care unit

HOMA-IR and HLA-DR expression on monocytes

Figure 1 presents the dynamic changes of the HOMA-IR score reflecting IR over time. There was no significant difference between the two groups with respect to HOMA-IR at baseline. In contrast, intensive insulin therapy decreased the HOMA-IR score on postoperative days 1, 3, and 5 (3.01 ± 0.78 vs. 3.51 ± 0.73, $P < 0.001$; 3.42 ± 0.77 vs. 3.84 ± 0.74, $P < 0.001$; 2.89 ± 0.70 vs. 3.65 ± 0.54, $P < 0.001$, respectively, for IG vs. CG). Figure 2 shows monocytes' HLA-DR expression (%) at different time points in all patients. Analysis revealed no significant effect on HLA-DR expression at baseline and on postoperative day 1 between the two groups. However, it differed significantly on postoperative days 3 and 5 (54.3 ± 10.0 vs.

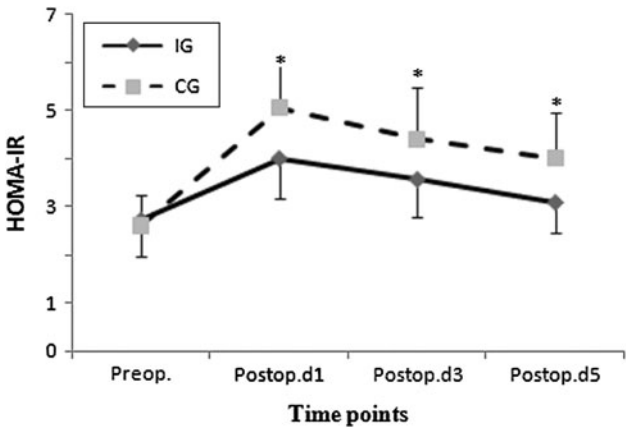

Fig. 1 HOMA-IR variation during perioperative period. HOMA-IR, insulin resistance of homeostasis model assessment; *Preop.* preoperative, *Postop.* postoperative, *IG* intensive group, *CG* conventional group. * $P < 0.01$ vs. conventional group

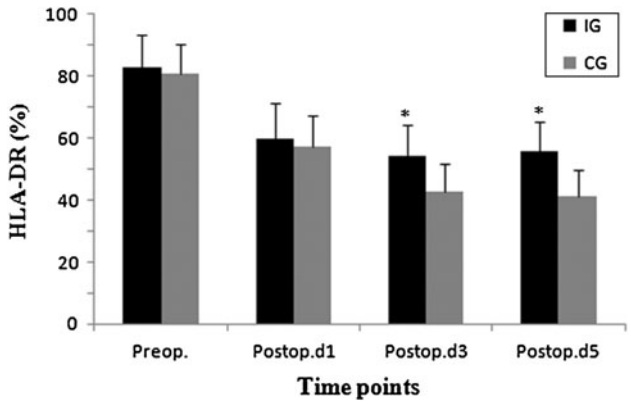

Fig. 2 The percentage of monocytes expressing HLA-DR during perioperative period. HLA-DR, human leukocyte antigen (HLA)-DR. * $P < 0.01$ vs. conventional group

42.8 ± 9.1, $P < 0.001$; 55.7 ± 9.6 vs. 41.1 ± 8.5, $P < 0.001$).

Discussion

Although D2 dissection is associated with excessive morbidity compared to D1 dissection, it is increasingly recommended for curative treatment of gastric cancer. DM is one of the major risk factors in surgical patients, contributing to the development of postoperative nosocomial infections. Yamashita et al. [18] found that patients with long-term DM had a significantly greater incidence of postoperative nosocomial infections after elective gastrectomy. In this study, we focused on a homogeneous population with type 2 DM who underwent elective radical gastrectomy. We showed that intensive insulin therapy with the goal of maintaining the blood glucose level between 4.4 and 6.1 mmol/l significantly reduced postoperative short-term morbidity. However, IIT had no significant effect on 28-day mortality and it increased the risk for hypoglycemia.

In 2006, a study revealed that IIT significantly reduced mortality and morbidity in mixed medical and surgical ICU patients, and this effect was pronounced when IIT was continued for at least 3 days [19]. By contrast, in 2008, a meta-analysis found that tight glucose control was not associated with a significant reduction in hospital mortality in critically ill adults and the significant association with reduced risk of septicemia was limited to trials conducted in surgical ICU [20]. Our findings showed that IIT did not significantly alter 28-day mortality among the type 2 DM patients who underwent D2 gastrectomy. However, the data showed that IIT reduced the risk of postoperative complications. Moreover, IIT shortened days to suture removal, postoperative hospital stay, and postoperative duration of antibiotic use in this population. Of note, since the study involved patients with type 2 DM who underwent elective radical gastrectomy, the results cannot be extrapolated to patients without DM or with type 1 DM or those who had other types of surgery who were not in the study population.

Although the multidisciplinary team performed the continuous insulin infusion protocol in a strict manner, the incidence of hypoglycemia could not be eliminated completely. In CG, 1.1% of patients versus 6.1% of patients in IG had severe hypoglycemia, defined as a blood glucose level of less than 2.2 mmol/l. There were no short- or long-term sequelae in these patients because of appropriate intervention. Van den Berghe et al. reported that IIT increased the risk of hypoglycemia from 0.8 to 5.2% in the surgical ICU Leuven study [4] and from 3.1 to 18.7% in the medial ICU Leuven study [6] with the same definition of hypoglycemia. Importantly, we found that hypoglycemia tended to occur during the early postoperative period. It was thought that the early signs of hypoglycemia may be masked by the administration of sedatives, analgesics, and anesthetics, which reminds us that we should look for the signs of hypoglycemia during this period. In addition, multidisciplinary education on IIT is vital to achieving an optimal glucose target and to ensure treatment safety in postoperative patients. Therefore, the risk of hypoglycemia should not be considered an obstacle to using intensive insulin therapy in this population.

Ways to improve of IIT remain unknown. Amelioration of insulin sensitivity [21–23], prevention of glucose toxicity [24], suppression of excessive inflammation [9, 25], and improvement of monocytes phagocytosis capacity [26] may partially explain the benefits of intensive insulin therapy. Hyperglycemia associated with IR is common in critically ill patients, especially in those who previously had DM. In patients who undergo moderate to severe surgery, postoperative IR can be overcome if a sufficient amount of insulin is infused to maintain normoglycemia [27]. Few studies have attempted to detect whether IIT influences the postoperative IR in type 2 DM patients. As illustrated in Fig. 2, although the postoperative HOMA-IR score in all patients increased compared to baseline, IIT significantly decreased the HOMA-IR score compared to CIT. This implies that IIT suppressed the postoperative IR in this population.

The effect of IIT on HLA-DR remains unclear. Importantly, it has been demonstrated that there is an association between low HLA-DR expression and impairment of various monocytic functions; the restoration of normal HLA-DR levels is associated with the recovery of monocytic functions. Up to now, the decrease of HLA-DR expression has been assessed as mainly a predictor of septic complications after trauma, surgery, or pancreatitis [28–30]. In this study, the decrease of postoperative HLA-DR expression in both groups indicated that monocytic functions were impaired. Simultaneously, the results suggested that IIT improved the postoperative monocytic dysfunction.

It is noteworthy that laparoscopic gastrectomy with lymph node dissection has developed as a minimally invasive surgical procedure for gastric cancer over the past decade. Laparoscopic gastrectomy has advantages over open surgery, including faster recovery, earlier resumption of gastrointestinal function, attenuated immune depression, and shorter hospital stay. We speculate that it is equally important to maintain an appropriate level of perioperative blood glucose for type 2 DM patients undergoing laparoscopic gastrectomy [31, 32]. However, additional studies focusing on this patient population should be designed to verify whether it is necessary to implement intensive insulin therapy and whether the same findings would be found with a laparoscopic gastrectomy cohort.

The current study has certain limitations. This was a single-center study. It was not strictly blinded because the infusion of insulin, monitoring of blood glucose levels, and appropriate hypoglycemia intervention needed to be performed by the IIT team. However, to minimize the bias, the staff responsible for adjustment of insulin dose and monitoring of blood glucose levels was unaware of clinical decision-making and important outcome measurements. Finally, the long-term effect of intensive insulin therapy was not investigated in this study. However, the study population was more homogeneous compared to other studies, which made the results more powerful.

In conclusion, although maintenance of normoglycemia with IIT during the early postoperative period did not decrease mortality, it reduced morbidity among type 2 DM patients who underwent D2 gastrectomy. Furthermore, suppression of the IR and improvement of HLA-DR expression may partially mediate the advantageous effects of IIT. Moreover, IIT increased the risk of hypoglycemia, but brief hypoglycemia, diagnosed and treated promptly and exactly, did not cause severe short- or long-term sequelae. Large, multicenter, blinded trials are needed to confirm these preliminary results.

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Conflict of interest There is no potential and real conflict to any third party.

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